SALES DATA ANALYSIS USING PYTHON

Python | Pandas | NumPy | Matplotlib

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TOOLS & TECHNOLOGIES USED

- **Python**

1. –Pandas
2. –NumPy
3. –Matplotlib
4. –Jupyter Notebook

PANDAS

```
import pandas as pd
```

•First 20 rows

	OrderID	Date	Region	Product	Category	Units_Sold	Unit_Price	Discount (%)	Total_Sales	Discount_Amount	Final_Sales	Month
0	1001	2025-01-01	East	Laptop	Electronics	5	800	10	4000	400.0	3600.0	2025-01
1	1002	2025-01-02	West	Phone	Electronics	10	500	5	5000	250.0	4750.0	2025-01
2	1003	2025-01-03	North	Chair	Furniture	8	120	0	960	0.0	960.0	2025-01
3	1004	2025-01-04	South	Table	Furniture	3	300	15	900	135.0	765.0	2025-01
4	1005	2025-01-05	East	Headphones	Electronics	15	100	5	1500	75.0	1425.0	2025-01
5	1006	2025-01-06	West	Monitor	Electronics	7	250	8	1750	140.0	1610.0	2025-01
6	1007	2025-01-07	North	Sofa	Furniture	2	900	12	1800	216.0	1584.0	2025-01
7	1008	2025-01-08	South	Keyboard	Electronics	20	40	5	800	40.0	760.0	2025-01
8	1009	2025-01-09	East	Mouse	Electronics	25	25	0	625	0.0	625.0	2025-01
9	1010	2025-01-10	West	Desk	Furniture	4	450	10	1800	180.0	1620.0	2025-01
10	1011	2025-01-11	North	Tablet	Electronics	6	300	7	1800	126.0	1674.0	2025-01
11	1012	2025-01-12	South	Printer	Electronics	3	200	5	600	30.0	570.0	2025-01
12	1013	2025-01-13	East	Bookshelf	Furniture	5	150	10	750	75.0	675.0	2025-01
13	1014	2025-01-14	West	Smartwatch	Electronics	9	220	6	1980	118.8	1861.2	2025-01
14	1015	2025-01-15	North	Bed	Furniture	1	1200	15	1200	180.0	1020.0	2025-01
15	1016	2025-01-16	South	Laptop	Electronics	4	850	10	3400	340.0	3060.0	2025-01
16	1017	2025-01-17	East	Phone	Electronics	12	520	5	6240	312.0	5928.0	2025-01
17	1018	2025-01-18	West	Chair	Furniture	10	110	0	1100	0.0	1100.0	2025-01
18	1019	2025-01-19	North	Table	Furniture	6	320	10	1920	192.0	1728.0	2025-01
19	1020	2025-01-20	South	Headphones	Electronics	18	90	5	1620	81.0	1539.0	2025-01

TOTAL_SALES = UNITS_SOLD × UNIT_PRICE

```
df["Total_Sales"] = df["Units_Sold"] * df["Unit_Price"]
```

```
df
```

	OrderID	Date	Region	Product	Category	Units_Sold	Unit_Price	Discount (%)	Total_Sales
0	1001	2025-01-01	East	Laptop	Electronics	5	800	10	4000
1	1002	2025-01-02	West	Phone	Electronics	10	500	5	5000
2	1003	2025-01-03	North	Chair	Furniture	8	120	0	960
3	1004	2025-01-04	South	Table	Furniture	3	300	15	900
4	1005	2025-01-05	East	Headphones	Electronics	15	100	5	1500
...	...	...	...	...	...	...	...	...	...
100	1101	2025-04-11	East	Tablet	Electronics	11	335	7	3685
101	1102	2025-04-12	West	Printer	Electronics	6	240	5	1440
102	1103	2025-04-13	North	Bookshelf	Furniture	4	200	10	800

#DISCOUNT_AMOUNT

```
df["Discount_Amount"] = (df["Units_Sold"] * df["Unit_Price"]) * (df["Discount (%)"] / 100)
```

```
df
```

	OrderID	Date	Region	Product	Category	Units_Sold	Unit_Price	Discount (%)	Total_Sales	Discount_Amount
0	1001	2025-01-01	East	Laptop	Electronics	5	800	10	4000	400.00
1	1002	2025-01-02	West	Phone	Electronics	10	500	5	5000	250.00
2	1003	2025-01-03	North	Chair	Furniture	8	120	0	960	0.00
3	1004	2025-01-04	South	Table	Furniture	3	300	15	900	135.00
4	1005	2025-01-05	East	Headphones	Electronics	15	100	5	1500	75.00
...	...	...	...	...	...	...	...	...	...	...
100	1101	2025-04-11	East	Tablet	Electronics	11	335	7	3685	257.95
101	1102	2025-04-12	West	Printer	Electronics	6	240	5	1440	72.00
102	1103	2025-04-13	North	Bookshelf	Furniture	4	200	10	800	80.00
103	1104	2025-04-14	South	Smartwatch	Electronics	15	255	6	3825	229.50
104	1105	2025-04-15	East	Bed	Furniture	1	1400	15	1400	210.00

#FINAL_SALES = TOTAL_SALES - DISCOUNT

```
df["Final_Sales"] = df["Total_Sales"] - df["Discount_Amount"]
```

```
df
```

	OrderID	Date	Region	Product	Category	Units_Sold	Unit_Price	Discount (%)	Total_Sales	Discount_Amount	Final_Sales
0	1001	2025-01-01	East	Laptop	Electronics	5	800	10	4000	400.00	3600.00
1	1002	2025-01-02	West	Phone	Electronics	10	500	5	5000	250.00	4750.00
2	1003	2025-01-03	North	Chair	Furniture	8	120	0	960	0.00	960.00
3	1004	2025-01-04	South	Table	Furniture	3	300	15	900	135.00	765.00
4	1005	2025-01-05	East	Headphones	Electronics	15	100	5	1500	75.00	1425.00
...	...	...	...	...	...	...	...	...	...	...	...
100	1101	2025-04-11	East	Tablet	Electronics	11	335	7	3685	257.95	3427.05
101	1102	2025-04-12	West	Printer	Electronics	6	240	5	1440	72.00	1368.00
102	1103	2025-04-13	North	Bookshelf	Furniture	4	200	10	800	80.00	720.00
103	1104	2025-04-14	South	Smartwatch	Electronics	15	255	6	3825	229.50	3595.50
104	1105	2025-04-15	East	Bed	Furniture	1	1400	15	1400	210.00	1190.00

WHAT IS THE TOTAL REVENUE?

```
Total_revenue = df["Final_Sales"].sum()
print(Total_revenue)
```

```
201513.65000000002
```

#WHICH REGION HAS HIGHEST SALES?

```
region_sales = df.groupby('Region')['Final_Sales'].sum().sort_values(ascending=False)
print(region_sales)
print("Highest Sales Region:", region_sales.idxmax())
```

```
Region
West     53938.30
South    52256.55
East     51716.10
North    43602.70
Name: Final_Sales, dtype: float64
Highest Sales Region: West
```

#WHICH CATEGORY PERFORMS BEST?

```python
category_sales = df.groupby('Category')['Final_Sales'].sum().sort_values(ascending=False)
print(category_sales)
print("Best Category Performs:", category_sales.idxmax())
```

```
Category
Electronics    147162.85
Furniture       54350.80
Name: Final_Sales, dtype: float64
Best Category Performs: Electronics
```

#WHAT IS THE AVERAGE DISCOUNT?

```python
avg_discount = df['Discount (%)'].mean()
print("Average Discount (%):", avg_discount)
```

```
Average Discount (%): 7.485714285714286
```

#Which product generates highest profit?

```python
product_profit = df.groupby('Product')['Discount_Amount'].sum().sort_values(ascending=False)
print(product_profit)

print("Highest Profit Product:", product_profit.idxmax())
```

```
Product
Laptop        2671.00
Phone         2476.00
Sofa          1747.20
Bed           1704.00
Table         1555.50
Tablet        1256.15
Smartwatch    1196.40
Desk          1038.00
Monitor        999.60
Headphones     650.10
Bookshelf      575.50
Printer        361.25
Keyboard       294.65
Chair            0.00
Mouse            0.00
Name: Discount_Amount, dtype: float64
Highest Profit Product: Laptop
```

#Correlation between discount and units sold?

```
correlation = df['Discount (%)'].corr(df['Units_Sold'])
print("Correlation between Discount and Units Sold:", correlation)
```

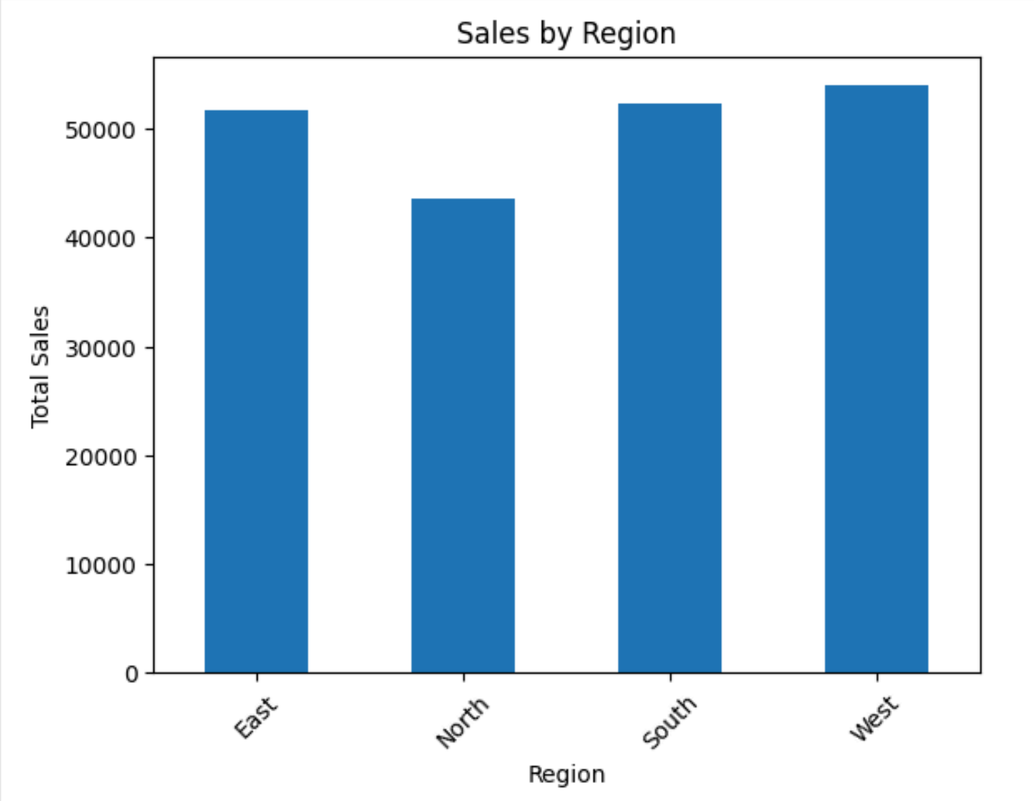
```
Correlation between Discount and Units Sold: -0.7231202492682444
```

#VISUALIZATION TASKS (MATPLOTLIB)

```
import matplotlib.pyplot as plt
```

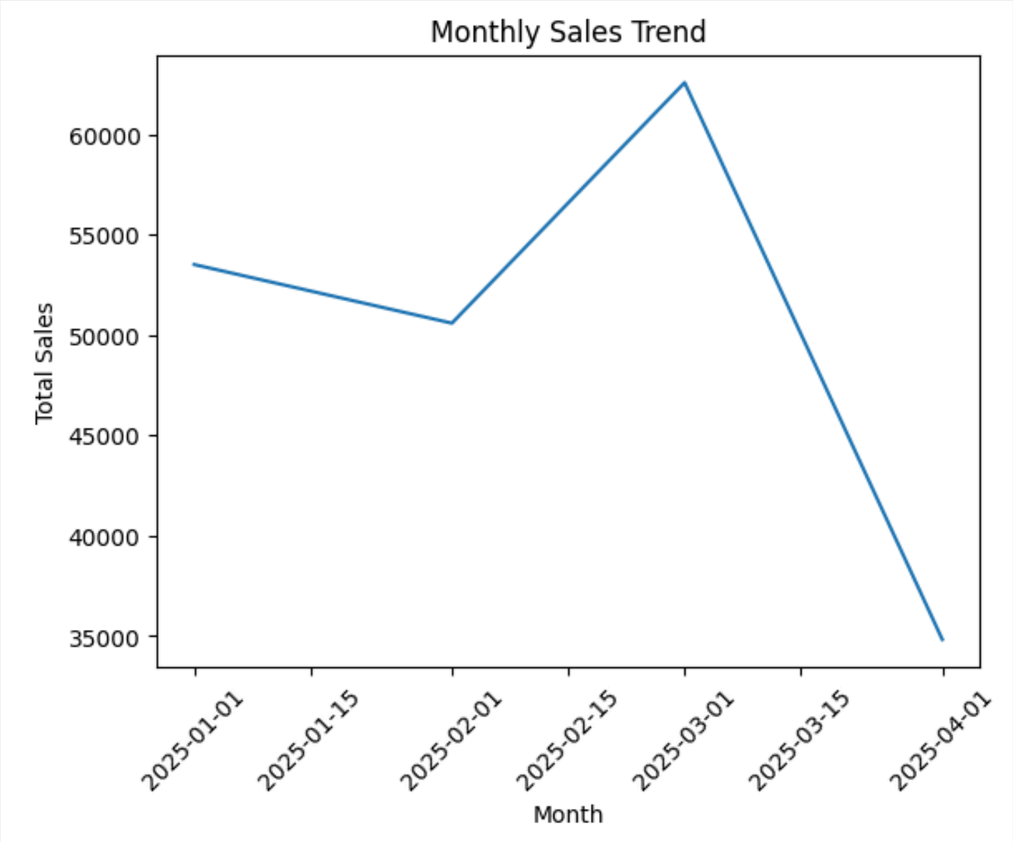
#Bar chart → Sales by Region

```
region_sales = df.groupby('Region')['Final_Sales'].sum()
plt.figure()
region_sales.plot(kind='bar')

plt.title('Sales by Region')
plt.xlabel('Region')
plt.ylabel('Total Sales')
plt.xticks(rotation=45)
plt.show()
```



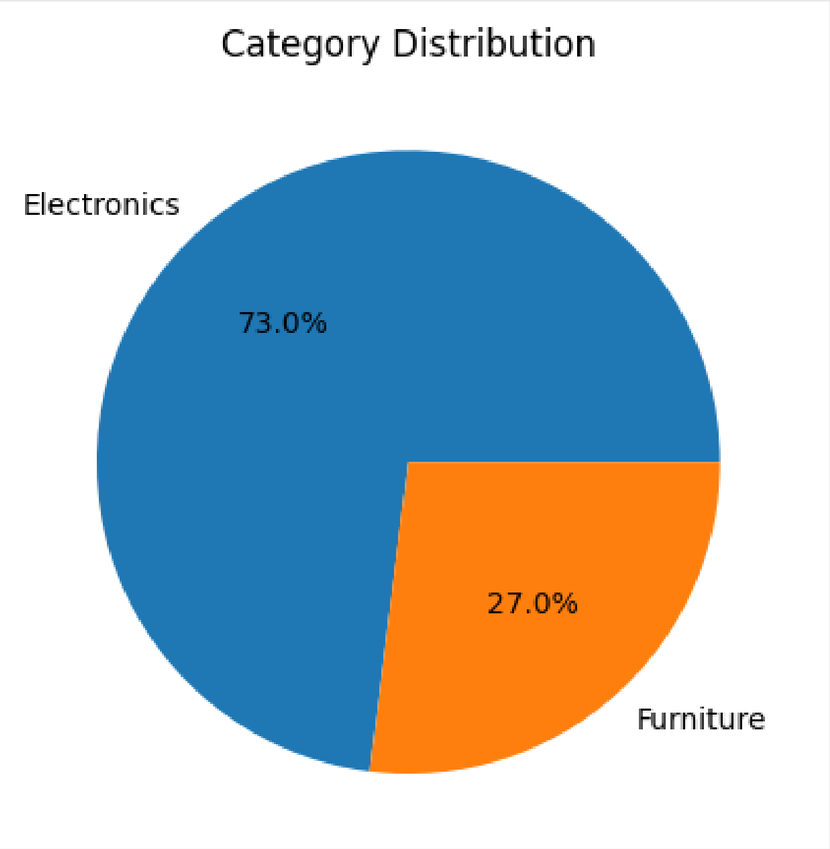
#Line chart → Sales trend over time

```
df['Month'] = df['Date'].dt.to_period('M')
monthly_sales = df.groupby('Month')['Final_Sales'].sum()
monthly_sales.index = monthly_sales.index.to_timestamp()

plt.figure()
plt.plot(monthly_sales.index, monthly_sales.values)

plt.title('Monthly Sales Trend')
plt.xlabel('Month')
plt.ylabel('Total Sales')
plt.xticks(rotation=45)
plt.show()
```



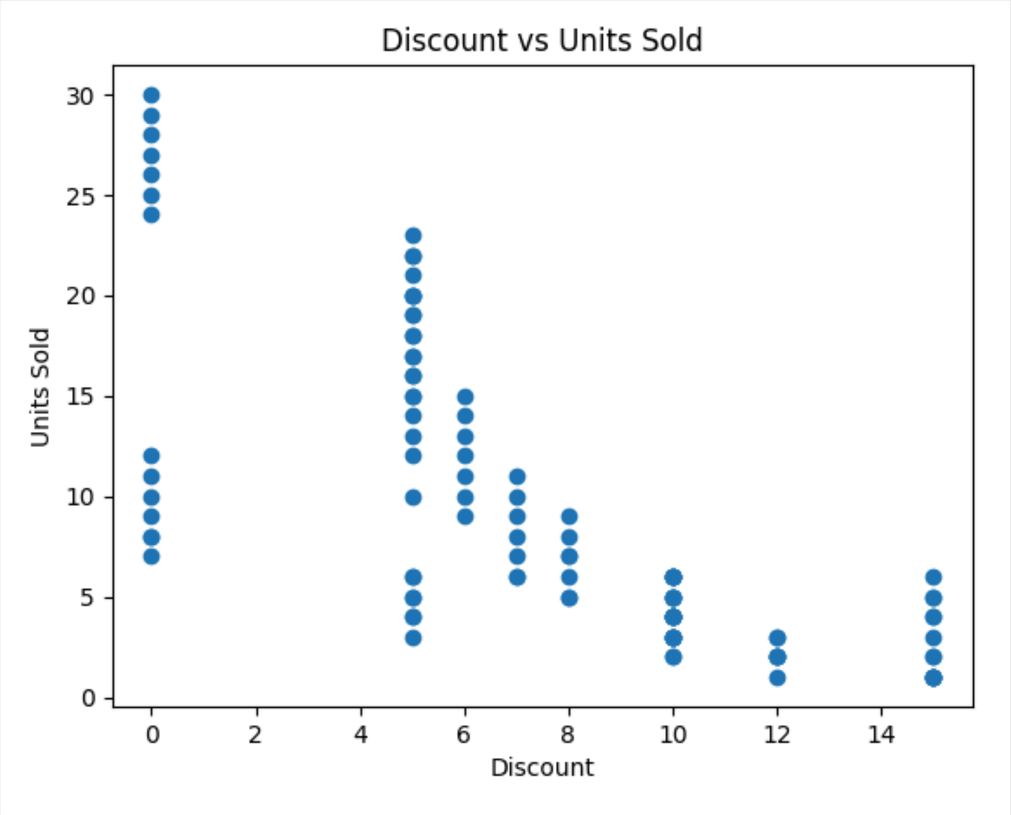
#Pie chart → Category distribution

```
category_sales = df.groupby('Category')['Final_Sales'].sum()


plt.figure()
plt.pie(category_sales, labels=category_sales.index, autopct='%1.1f%%')

plt.title('Category Distribution')
plt.show()
```



#Scatter plot → Discount vs Units Sold

```
plt.figure()
plt.scatter(df['Discount (%)'], df['Units_Sold'])

plt.title('Discount vs Units Sold')
plt.xlabel('Discount')
plt.ylabel('Units Sold')
plt.show()
```



NumPy

```
import numpy as np
```

#Calculate:
#Mean sales
#Standard deviation
#Variance

```
sales = df['Final_Sales']

mean_sales = np.mean(sales)
std_sales = np.std(sales)
variance_sales = np.var(sales)

print("Mean Sales:", mean_sales)
print("Standard Deviation:", std_sales)
print("Variance:", variance_sales)
```

```
Mean Sales: 1919.1776190476194
Standard Deviation: 1559.218549313374
Variance: 2431162.4845229024
```

THANK YOU